MODEL REFERENCE

A one-dimensional radiative-convective atmospheric column

18,450 lines of C# that march a 50 km column of air to radiative–convective equilibrium, with longwave opacity built from real HITRAN line data rather than a tuned grey constant — and with the far wings of CO_2's 15 µm band corrected for the sub-Lorentzian shape that sets its forcing.

What it computes

Given a solar input, a planetary albedo and an absorber loading, it solves for the temperature at every level of an atmospheric column and at the surface beneath it — the state in which the column radiates to space exactly what it absorbs, and no level is left with a net heating tendency.

Two balances have to close at once, and they are not independent. At the top of the atmosphere the outgoing longwave must equal the absorbed sunlight. At the surface, the absorbed sunlight plus the back radiation must equal what the ground emits plus what convection and evaporation carry away. A column can satisfy either alone; equilibrium is where both hold together.

SURFACE TEMPERATURE	OUTGOING LONGWAVE	GREENHOUSE EFFECT	CONVECTING LAYER
286.797 K	238.175 W m⁻²	137.8 W m⁻²	3.44 km
13.65 °C, default configuration	equal to absorbed solar at equilibrium	surface emission minus outgoing longwave	above it, radiation alone

Everything below is how those numbers are arrived at, and — equally — which of them are predictions and which are calibrations. That distinction is kept explicit throughout, because a model of this kind can be made to produce almost any answer if the difference is allowed to blur.

The vertical grid

The column is divided into segments of equal geometric thickness from the surface to 50 km. Each carries a pressure at its lower and upper interface, a mid-layer pressure, and an air mass per unit horizontal area obtained from hydrostatic balance:

$$dp = -\rho\ g\ dz$$

so the mass supporting a pressure drop is *Δp/g*, independent of the density profile

Pressures come from the US Standard Atmosphere. That standard is defined on *geopotential* altitude, which measures height in units of work done against a constant reference gravity, not in metres. The two differ by

$$H = r_0 z / (r_0 + z)$$

about 140 m at 50 km on Earth

and the model converts between them rather than treating the standard's tabulated heights as geometric. Ignoring the distinction would put the top of the column in the wrong place by more than the thickness of a segment.

Every flux in the model is **power per unit area of planet surface**, including inside the atmosphere. That convention is what makes the energy budgets close arithmetically, and it is what allows spherical geometry to be added later as a single scaling rather than a rewrite.

Longwave emission: the Koenigsberger equation

A volume of gas at temperature T emits

$$dq = 4\ \varepsilon'\ \sigma\ T^4\ dV$$

ε' is a volumetric emission coefficient, m^{-1}

The factor of four is not decorative: it is the ratio of the surface area of a sphere to the area of its shadow, and it is what converts a directional emission into an isotropic one. Because every quantity here is per unit horizontal area, dV reduces to a thickness dz, and a segment's total emission is $4\ \varepsilon'\ \sigma\ T^4\ dz$.

This is the optically thin limit. The solver does not use it directly — it uses the exponential form below, which reduces to exactly this as $dz \to 0$, and the model checks that it does.

Why the diffusivity factor is exactly two

Radiation crosses a slab in every direction, not just vertically. A ray at zenith angle θ travels a path 1/cos θ longer than the vertical one, so a flux calculation has to integrate over the hemisphere. That integral gives the exponential integral E_3, and the transmittance of a slab of vertical optical depth τ is $2E_3(\tau)$.

Two-stream models replace that integral with a single effective path, *D* times the vertical one. The usual choices are *D* = 1.66 (Elsasser) or 2. This model takes *D* = 2, and the reason is that it is the only value consistent with the emission law above:

$$2E_3(\tau) \to 1 - 2\tau \quad \text{as } \tau \to 0$$

so absorptivity → 2τ, matching the $4\varepsilon'\sigma T^4$ emission exactly

Choosing *D* = 1.66 would make the column absorb 17 % less than it emits in the thin limit, which violates Kirchhoff's law at the level of the model's own construction. The better far-field accuracy of 1.66 is real, but it is bought by breaking an identity the rest of the model depends on. *D* is a parameter; the default is the consistent one.

The two-stream recurrence

Upward and downward fluxes are marched across segments. For a segment of hemispheric optical thickness $\tau = D\,\varepsilon'\,dz$, with transmittance $t = e^{-\tau}$:

$$F_{i+1}^{\uparrow} = t\,F_i^{\uparrow} + (1 - t)\,\sigma T_i^4$$

and symmetrically downward, from the top

This is exact, not a discretisation. For a segment at constant temperature the Schwarzschild equation integrates to precisely this expression, which has a consequence worth stating: subdividing an isothermal slab leaves its emission unchanged. Grid refinement in this model therefore tests the *temperature structure*, not the radiative transfer, and any convergence error that shows up is about where the temperature is sampled rather than about how radiation is propagated.

The boundary conditions are that nothing enters the top of the column, and that the surface emits $\varepsilon_s\sigma T_s^4$ and reflects the fraction $(1 - \varepsilon_s)$ of the back radiation reaching it.

From a grey slab to real spectroscopy

A grey model gives every wavelength the same opacity. Real gases do not: CO_2 is nearly opaque at 15 μm and transparent at 10, and that difference is the whole reason the response to more CO_2 is logarithmic rather than linear.

The model can be run either way. In its spectral mode it takes HITRAN line lists for six absorbers — water vapour in two bands, CO_2's 15 μm band, ozone at 9.6 μm, methane at 7.7 μm and nitrous oxide at 7.8 μm — and derives bands from the line data itself:

1. **Resolve the spectrum.** Each line is given a Lorentz profile with half-width $\gamma = \gamma_{ref}(p/p_{ref})(296/T)^n$, where *n* is HITRAN's own temperature exponent. Collisions are more frequent at higher pressure and faster at higher temperature, and the two pull the width in opposite directions — a stratospheric layer at a tenth of an atmosphere and 220 K comes out about 12 % broader than pressure alone would give.
2. **Split into bands.** Band edges are placed so each carries a comparable share of the Planck function at a representative temperature, rather than being equally spaced in wavenumber.
3. **Measure a k-distribution per band.** The resolved absorption coefficients inside a band are sorted and reduced to a small number of quadrature points — the correlated-k method. Sorting is what makes this work: the reordered spectrum is smooth, so a handful of points integrates it accurately where thousands would be needed in wavenumber order.

All six gases are derived on one shared wavenumber grid. Deriving them separately would produce overlapping band sets — N_2O's 7.8 μm band sits inside methane's, and water vapour is everywhere — and overlapping bands each claim their own share of every emitter's Planck function, so the spectrum would sum to more than one.

The far wings, and the correction that changed the answer

The Lorentz profile comes from the impact approximation: collisions treated as instantaneous compared with the time between them. That holds near line centre and fails far from it. At a detuning of tens of cm^{-1} the radiation is probing the collision itself, over times short enough that its finite duration matters, and real CO_2 wings fall off **faster** than Lorentzian.

That is not a detail here. This model's CO_2 forcing comes almost entirely from the far wings — the band core is saturated, so only the wings can still respond to more gas — which is precisely *why* the response is logarithmic. The far-wing shape is the thing that sets the forcing coefficient.

The evidence arrived before the explanation. A convergence study over the wing cutoff gave a forcing coefficient of 5.15, 6.63, 6.88 and 6.98 W m^{-2} per ln as the cutoff opened from 100 to 800 cm^{-1} — the answer was being set by how far out the wings were integrated, and it converged on a value about 1.30 times the accepted 5.35.

The correction is the three-segment exponential of Perrin & Hartmann (1989), multiplying the Lorentz profile:

$$\chi(s) = \exp[\ -B_1(s-s_1) - B_2(s-s_2)^+ - B_3(s-s_3)^+\ ]$$

$s = |\nu - \nu_0|$, breakpoints at 3, 50 and 180 cm^{-1}, continuous at each

It is applied to **CO_2 alone**. A χ factor is fitted to one band of one gas; giving CO_2's to water vapour would be inventing spectroscopy rather than correcting it. It is also applied *outside* the k-distribution, because droplet-free line absorption and the χ correction are different functions of wavenumber and folding them together would misplace the correction.

	ABSORBER SCALE	BASE T_S	A	VS ACCEPTED
Pure Lorentz wings	14.578	286.796 K	6.988	1.31×
χ factor, same loading	14.578	285.109 K	5.073	0.95×
χ factor, recalibrated	15.869	286.796 K	5.091	0.95×

Measured in two steps, so the shape change and the loading change are not confounded

Essentially the entire over-forcing was that one piece of missing spectroscopy. Two things are worth reading off the table. Suppressing the wings removes absorption, so the base state cooled 1.7 K and the gas loading had to come back up — but **recalibrating barely moved the coefficient**, 5.073 to 5.091, which is the reassurance that it is a property of the spectroscopy and not of how much gas was put in.

Those absorber scales are the ones this experiment was run at, not the shipped values, and the coefficients above predate both the methane calibration and the correction to the χ coefficients described below. The table is left as measured because its point is the *controlled comparison* between three states; restating it would mean re-running all three rather than editing one column. The current numbers are in the results table at the end.

One thing the loading column does *not* explain, and it is worth saying because it is the obvious suspicion. Raising the absorber scale does not push the coefficient down: measured directly, 15.869 gives 4.956 and 17.762 gives 4.975, and halving the loading to 8.881 *lowers* it to 4.788. (Those three were measured together before the χ correction below and with a through-origin fit, so they are comparable with each other rather than with the results table; the point is the ordering, which the correction does not touch.) Extra overlying water vapour masking CO_2 is a real mechanism — it is what the unfitted configuration below demonstrates, where vapour carries 98.8 % of the optical depth against CO_2's 1.2 % — but at this recipe's four-to-one ratio it is not what sets the coefficient.

THE COEFFICIENTS WERE CHECKED AGAINST THE LITERATURE, AND WERE WRONG

The paragraph that used to sit here said the coefficients were representative values that had not been checked against the original paper. They have now been checked, and **they were wrong in two separate ways.**

The citation was to the wrong band. Perrin & Hartmann (1989) — *JQSRT* **42**(4), 311–317 — measures the far wings of the **4.3 μm ν_3** band of CO_2 in N_2. It is not a 15 μm ν_2 result at all. The three-segment functional form above is general and is genuinely theirs, but the coefficients are fitted per band *and* per collision partner, and one band's numbers are not transferable to another. Oxford's RFM, a production 15 μm line-by-line model, cites Cousin *et al.* (1985) for its χ factor rather than Perrin & Hartmann — the same point seen from the other side.

The numbers themselves matched no published set. The current tabulation is Chaverot *et al.* (2025), *A&A* **702**, A137, whose author list includes Hartmann and which is an explicit update of the older factors. Its Table 1 confirms the functional form used here is exactly right. Its Tables 2 and 3 also show that the B coefficients are **temperature dependent**, $B_i(T) = \alpha_i + \beta_i \exp(-\gamma_i T)$, for which this model previously had no term at all.

The relevant column is CO_2–N_2, ν_2: the absorber here is CO_2 at trace concentration in a bath of nitrogen, so nitrogen is what broadens its lines.

	WAS	PUBLISHED
s_1 / s_2 / s_3 (cm^{-1})	3 / 30 / 120	3 / 50 / 180
B_1	0.0888	0.0806
B_2	0.0232	0.0181
B_3	0.0160	**0.0085**
temperature dependence	none	$\alpha + \beta\ e^{-\gamma T}$

What the model carried, against the measured CO_2–N_2 ν_2 values at 296 K

The error ran in both directions, which is why guessing its sign would have been wrong. Because s_2 was 30 rather than 50, the steep first decay stopped twenty cm^{-1} too early and the gentler second segment took over too soon: the old χ was **2.5× too weak** at 50 cm^{-1} and stayed about twice too weak out to 250. Past roughly 300 cm^{-1} the doubled B_3 took over in the other direction and it became too *strong* — 0.39× the published value at 400 cm^{-1}. Both errors sat inside the wing cutoff, in the region that supplies the logarithm.

And correcting them barely moved the answer. The coefficient went from 4.975 to 4.958 W m^{-2} per ln — the two errors very nearly cancelled. That is worth stating plainly because the obvious hypothesis, that unverified coefficients were over-suppressing the wings and causing the model to under-force, is *false*. The correction was worth making because it is right, not because it changed the result.

One simplification remains, now visible rather than hidden. The published χ is applied per collision partner and weighted by that partner's share of the line width; this model applies a single χ. For an atmosphere whose broadening is overwhelmingly by air that is the right approximation, but it is an approximation.

WHAT THE SHORTFALL ACTUALLY IS: THE WING CUTOFF

With the coefficients right, the remaining gap has a different and simpler cause. The corrected far wing decays as $e^{-0.0085\,s}$, an e-folding length of 118 cm^{-1}, where the old one fell off in 62 cm^{-1}. A gentler wing reaches further, so the shipped 400 cm^{-1} cutoff truncates more of it than it used to.

WING CUTOFF (CM^{-1})	A	VS ACCEPTED	BASE T_S
400 (shipped)	4.958	0.927×	286.781 K
800	5.156	0.964×	287.068 K
1600	5.171	0.966×	287.109 K

Corrected χ, absorber scale 17.762, A fitted by least squares through the origin over all ten concentrations

It converges by 800 cm^{-1} — the last doubling moves A by 0.3 % — on **0.966× the accepted law, within 3.4 %**. The shipped 400 cm^{-1} cutoff is therefore no longer converged, and about four fifths of the apparent shortfall is that truncation rather than anything spectroscopic. The cutoff is left at 400 because it is what every other number in this document was measured at; changing it would mean re-running all of them, and the honest thing in the meantime is to say what it costs.

Why the response is logarithmic

Doubling the CO_2 doubles the absorber, but does not double the forcing. The reason is spectral saturation, and it can be seen in one line of algebra.

At the centre of the 15 μm band the atmosphere is already opaque: adding gas there changes nothing, because no more radiation can be stopped. The only place extra CO_2 can act is in the wings, where the band is still partly transparent. So the forcing is set by how fast the *opaque width* of the band grows.

If absorption falls off exponentially with distance from band centre, $k(\nu) \approx k_0 e^{-|\nu-\nu_0|/a}$, the band is opaque wherever $k\,u > 1$, which happens out to

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W = 2a ln(k₀ u)
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so $F \propto \ln u$ — the logarithm, in one step

The shape of the wing is doing all the work, and a different wing gives a different law. A Lorentzian wing, falling as $1/(\nu-\nu_0)^2$, would give $W \propto \sqrt{u}$ and hence a square-root response. A Gaussian wing would give $\sqrt{(\ln u)}$. Only something close to exponential gives a logarithm.

Which leads to a result worth stating plainly: correcting the wings made the model less logarithmic, not more. Drift of the fitted coefficient across the sweep went from 2–5 % to 8–10 %. That is not a regression — it follows directly from the algebra above. The pure Lorentzian far wing, integrated to a wide cutoff, happens to produce a very clean logarithm; suppressing it with a χ factor breaks that idealisation. The model was more logarithmic when it was more wrong.

The drift is a sign flip, not a magnitude. Reading the fitted coefficient at each concentration rather than its range says which way the departure runs, and χ reverses it. Without χ the coefficient *rises* across the sweep, 6.97 to 7.37: a Lorentzian wing decays slowly, so as the core saturates the band keeps expanding into wing that is still thin, and the reachable area grows faster than logarithmically. With χ it *falls*, 5.39 to 4.85 — the wing is cut back, so once the reachable part saturates there is less fresh spectrum left and each doubling buys less than the last.

Two consequences follow, and both were checked. Widening the wing cutoff hands the band more unsaturated wing, so it should drift less: 400 → 800 cm^{-1} takes the drift from 10.1 % to 8.2 %, which is the prediction made before the measurement rather than after it. And because the two tendencies have opposite sign, **a pure logarithm lies between them** — the model straddles the accepted law rather than missing it in one direction.

> One candidate was tested and ruled out. Re-deriving the bands at every concentration — the fix that rescued methane's response, where scaling a band's share stretched its mean while its k-distribution went on describing the reference atmosphere — does almost nothing here: 10.1 % becomes 10.0 %. It lowers the coefficient, from 4.717 to 4.534, but leaves the shape alone. Whatever sets CO_2's drift, it is not that approximation.
>
> The grey configuration makes the contrast concrete. Given one opacity for the whole spectrum, its forcing ratio against the accepted law grows from 1.17 to 2.14 across the same concentration span, because its absorber is simply diluted rather than spectrally resolved. There is no band core to saturate, so there is no logarithm to recover.

Methane, and why it obeys a different law

CO_2 is not the only gas the model can dial. Methane can be raised above its base value too, and it is a sharper test of the band machinery than CO_2 was — because it should obey a *different law*, and nothing in the model knows that.

The law follows the band's saturation, and the algebra of the previous section gives all three cases:

- **Optically thick** — the core is saturated, only the wings respond, and $F \propto \ln(C)$. This is CO_2's 15 μm band.
- **Partly saturated** — extra gas pushes more of the band from transparent to opaque, and $F \propto \sqrt{M}$. This is methane's 7.7 μm band, and it is the observed law: $0.036(\sqrt{M} - \sqrt{M_0})$ with M in ppb, from Myhre et al. (1998).
- **Optically thin** — nothing is blocking anything yet, absorption is simply proportional to amount, and $F \propto M$.

So a model that produces a logarithm for CO_2 and a square root for methane, out of the same machinery and without being told which is which, is demonstrating that the band structure carries real information. That is what it does — but only after two mistakes were corrected, and both are worth recording.

THE AMOUNT WAS NEVER CHECKED

Methane's share of the absorber recipe was 0.2, chosen to sit sensibly among the other gases and never compared with anything. At that loading the pre-industrial to present-day rise forced 3.42 W m^{-2} against an accepted 0.62 — five and a half times too much. The recipe sets *relative* amounts, not observed abundances.

Fixing it is a two-knob problem: the share sets the forcing, the absorber scale sets the base state, and cutting methane removes absorber. Solved together, three passes, holding the base state at 286.796 K.

THE SHAPE WAS HIDDEN BY AN APPROXIMATION

With the amount right, the response came out nearly **linear** — flatter than the observed square root, and over-predicting by 15 % at the top of the sweep.

> The obvious explanation was that the band had become optically thin. It is recorded here because it was wrong, and measuring it was what showed that: methane's hemispheric optical depth sits comfortably inside the partly-saturated range. Thinness was not what flattened the response.

The real cause was the approximation used to dial the gas. Scaling a band's recorded methane share stretches its *mean* optical depth while the k-distribution inside it goes on describing an atmosphere holding the reference amount — so the saturation that ought to develop in methane's strong lines never appears, and what is left is a small linear perturbation on a background the other gases are setting. The project's own notes warn about exactly this for CO_2, which already had a re-derivation path. Methane did not.

Re-deriving the bands at every concentration settles it:

	√M	LN M	LINEAR IN M
Scaled through the band share	0.0236	0.1202	**0.0033**
Re-derived per concentration	**0.0005**	0.0247	0.0457

Residual of each law's best fit through the origin — smallest wins

Re-derived and recalibrated, the square root beats a logarithm by fifty times and a straight line by ninety. The physics was in the line data throughout; the approximation was concealing it.

THE CLAIM THE CALIBRATION CANNOT MANUFACTURE

The share was fitted at 1900 ppb and nowhere else. At 3500 ppb — far outside the calibration — the model gives **0.976×** the accepted forcing, within 2.4 %. Before re-derivation that figure was 1.15. A one-point calibration cannot produce agreement at the far end; only the shape can.

What this costs is worth stating precisely, because it differs from CO_2. Methane's forcing *magnitude* is calibrated in and is not a prediction. Its *shape* is predicted, and comes out right. CO_2 is the other way around: its absorber scale is set by the base state, so its forcing coefficient — 0.886× the accepted law — remains a genuine prediction.

Both gases together

CO_2 and methane have not risen independently, so a figure that moves one and holds the other answers a question nobody is asking. Sweeping them together is a different kind of calculation from everything above — **it depends on an assumption about the future, and the assumption does more work than the physics.**

The coupling used here is stated so it can be disagreed with. Below present day the observed path is used: 285 ppm / 700 ppb to 421 ppm / 1920 ppb, a historical ratio of about 9 ppb per ppm. Above it, today's trends — CO_2 near 2.4 ppm a year and methane near 8 ppb — give 3.3 ppb per ppm. Extrapolating today's ratio *backwards* would contradict what actually happened, which is why the two halves differ.

> Every point beyond 421 ppm is an extrapolation of a current rate over centuries, not a projection. Real methane has a decade-scale atmospheric lifetime and a sink that responds to its own concentration; nothing here models either. Read the shape, not the year it might correspond to.

CO_2	CH_4	BOTH	CO_2 ALONE	METHANE'S SHARE
350	1283	+0.995 K	+0.787 K	21 %
425	1933	+1.901 K	+1.500 K	21 %
580	2450	+3.114 K	+2.572 K	17 %
700	2850	+3.819 K	+3.176 K	17 %
1000	3850	**+5.104 K**	+4.238 K	17 %

Equilibrium warming from the pre-industrial base state, both gases rising

Methane contributes about a fifth of the early warming and a sixth by 1000 ppm — its share falls as its own band saturates faster than CO_2's, which is the square root against the logarithm showing up in a temperature rather than a forcing.

THE FORCINGS ADD EXACTLY

Measured separately and together, at every point of the sweep:

$$F(\text{both}) - [\,F(CO_2) + F(CH_4)\,] = 0.0000\ \text{W m}^{-2}$$

at all ten concentrations, to four decimal places

That is not a modelling convenience, it is a measurement, and it confirms independently what the band structure says: CO_2 occupies 581–751 cm^{-1} and methane 1111–2000, and by the 360 cm^{-1} of detuning between them the χ factor has suppressed CO_2's wings to 5×10^{-4} of their Lorentzian value. Pure Lorentzian wings would have overlapped and the sum would not have closed.

A related check, run for the same reason: raising the CO_2 background from 285 to 1000 ppm raises methane's own forcing by 8–9 %. Holding the temperature profile fixed and adding only the CO_2 changes it by *nothing*, so that 8–9 % is the warmer base state emitting more for the same methane to block, and not a spectral interaction.

> Not covered: N_2O sits in methane's bands and is held fixed throughout, so this establishes only that CO_2 does not overlap methane. The accepted methane law carries an explicit CH_4–N_2O overlap term that this model has no way to produce.

Water vapour, and its feedback

Water vapour is the largest greenhouse gas in the column and the only absorber whose amount follows the temperature. Its loading is scaled by Clausius–Clapeyron integrated at constant latent heat,

$$e_{sat}(T) = e_\circ \exp[(L/R_v)(1/T_\circ - 1/T)]$$

evaluated at the near-surface air temperature

and distributed with its own scale height rather than following air density. Because it is re-evaluated before every radiation solve, a warming column becomes more opaque as it warms, which is the water-vapour feedback.

The feedback can be switched off, and the comparison is instructive — but only if it is set up carefully. Freezing the vapour at its nominal loading gives a *different base state*, so the two runs would start 2 K apart and the comparison would measure that offset. The no-feedback configuration is therefore frozen at the feedback run's own base air temperature, so both begin at exactly 286.796 K.

Set up that way, the two produce **identical forcings** — to four decimal places — and different warmings. That identity is not a coincidence: instantaneous forcing is defined at held temperatures, so a feedback cannot act on it. It is the cleanest demonstration in the model that a feedback changes the response and not the forcing.

> The feedback is a demonstration of mechanism, not a quantitative estimate. It scales the whole vapour column by one Clausius–Clapeyron factor taken at the surface, with a fixed exponential shape, so it cannot represent upper-tropospheric humidity — which is where the outgoing longwave is actually set. And the prescribed lapse rate means nothing can offset it, where a real lapse-rate feedback would cancel perhaps a third of it.

Clouds

Clouds do two opposite things, and the model does both. They are off by default.

In the shortwave they reflect. The albedo splits into a clear part and a cloudy part, mixed by cloud fraction:

$$A = (1-f)\,A_{clear} + f\,A_{cloud}$$

0.155 and 0.361 over 67 % of the sky returns Earth's 0.293

That split is what stops the clouds being counted twice. Earth's 0.30 planetary albedo *already contains* them; a reflective deck added on top of it would reflect the same sunlight again.

In the longwave they trap. The deck is specified by the emissivity it should have as a whole, inverted to an optical thickness $\tau = -\ln(1-\varepsilon)$ and spread over the segments by their *overlap* with the cloud rather than their full thickness — so a 1.0–4.5 km deck keeps the same emissivity however the column is divided up. Its opacity is grey, and is added outside the k-distribution: droplets absorb across a band where gas absorbs in lines. Folding it inside would have made the cloud thin wherever the gas is transparent, which is the exact opposite of what a cloud does, since its longwave work happens mostly in the window.

Two skies, one atmosphere. The longwave is solved twice, with and without the deck, and the fluxes mixed by cloud fraction — the independent column approximation. The mixing is exact, since fluxes add. What is approximate is the premise: that the cloudy and clear parts of the sky exchange no radiation sideways. Both solves see the same temperatures, because there is one atmosphere and one surface under both skies.

	MODEL	CERES
Shortwave — reflected away	−46.96	−47.1
Longwave — trapped	+26.55	+26.2
Net	**−20.41**	**−20.9**

Cloud radiative effect, W m^{-2}, against the CERES satellite record

Only the longwave figure is a result; the shortwave one is arithmetic on the two albedos, and those were chosen to reproduce Earth's. Both are measured as CERES measures them — this planet against the same planet with the cloud removed and nothing else touched.

The calibration is separate, and finding out why was the interesting part. Switching clouds on over the shipped configuration warms the surface by about 13 K. Nothing is broken: that configuration's absorbers were scaled to reach an Earth-like temperature at an 0.30 albedo with *no cloud* — a planet carrying the clouds' reflection while having none of their greenhouse. The gas had been standing in for the cloud greenhouse all along, so adding a real deck supplies it twice. The cloudy configuration therefore halves the gas loading and hands the difference to the deck, arriving at the same surface by a different route.

A second result fell out of that. Running the four corners — each effect on and off — the two nearly cancel in temperature while emphatically not cancelling in flux: reflection cools by 12.45 K, trapping warms by 11.75 K, and the net is **−0.99 K** against a net flux effect of −20.41 W m^{-2}. A single sensitivity applied to that flux would have predicted twelve. The components have different efficacies here — 0.27 against 0.44 K per W m^{-2} — which is a property of this model's fixed lapse rate and prescribed deck rather than a claim about Earth. It does illustrate why cloud feedback is the largest remaining uncertainty in real climate sensitivity: a net number that small is a difference between two much larger ones.

Convection and the surface

Radiative equilibrium alone gives a surface far hotter than observed and a lapse rate steeper than the atmosphere can sustain. Three non-radiative terms fix that.

A critical-lapse-rate adjustment. Wherever the lapse rate exceeds 6.5 K km^{-1} the affected block is mixed back onto exactly that rate, conserving $\Sigma\, C_i T_i$. This is not a diffusion — it is an instantaneous restoration of the profile a convecting atmosphere would hold, and it is applied after every timestep.

A surface sensible heat flux, $Q = h_c(T_s - T_{air})$, with $h_c = 5.8 + 4.1v$. The air temperature is extrapolated to z = 0 from the two lowest segments rather than taken from the lowest one, which matters more than it sounds: using the lowest segment evaluates the air at *dz*/2 and makes the whole model first order in *dz*. The radiative recurrence is already exact for constant-temperature segments, so this extrapolation is what restores second-order convergence.

An optional latent heat flux — evaporation — through the bulk aerodynamic form, off by default. Its moisture-availability parameter β scales the *whole flux*, not the surface humidity; scaling the humidity instead makes β < 1 drive perpetual dew deposition and *warm* the surface, which is the opposite of what suppressing evaporation should do.

The surface is deliberately excluded from the lapse-rate adjustment. It exchanges heat with the air through the flux above and through radiation, and folding it into the convective mixing would let it equilibrate with the air instantly, removing the temperature difference that drives the flux in the first place.

Geometry, optionally

Three corrections take the column from a plane-parallel idealisation toward a real planet. Each is switchable, and their sizes are worth knowing because two of them nearly cancel.

Sphericity. A shell at radius *r* has $(r/r_0)^2$ times the area of the surface beneath it. Writing $G = (r/r_0)^2 F$ — the power crossing radius *r*, divided by the area beneath it — removes the geometric term from the spherical two-stream equations entirely, leaving the plane-parallel equation with its source scaled by $(r/r_0)^2$. So sphericity enters the solver in exactly one line, the recurrence stays exact, and every budget still closes in W per m^2 of surface. Effect: −0.016 K.

Gravity falling with height. $g(z) = g_0(r_0/(r_0+z))^2$, about 1.5 % weaker at 50 km, which changes the mass supporting each pressure drop. Effect: +0.012 K.

The top-of-atmosphere solar cross-section. The planet intercepts sunlight on a disc of radius $r_0 + z_{top}$, not r_0, and rays near the limb pass through a long slant path without ever reaching the ground. Effect: +0.309 K — an order of magnitude larger than the other two, and the only one that materially matters.

Solving it

The column is marched to equilibrium with an adaptive explicit forward-Euler step on each segment's heat capacity. The step is limited by two constraints: no level may move more than a fixed amount in one step, and the step must stay inside the explicit stability limit set by the local radiative relaxation time C/λ, where λ is the derivative of net loss with respect to temperature.

That second constraint is what makes the problem stiff. The thin upper segments have small heat capacities and short relaxation times, so they set a step much shorter than the lower column needs, and a single march takes some 44,000 steps. It is the dominant cost of the model.

Convergence is measured across the *complete* step, adjustment included. Inside a convecting block an individual segment's radiative tendency does not vanish at equilibrium — it is balanced by the convective adjustment — so the tendency alone is not a convergence measure.

A concentration sweep runs twenty of these marches. Every concentration above the reference is independent of the others — each builds its own column and measures its forcing against the one shared reference state — so they run concurrently, and the sweep the charts need went from 217 seconds to 59 on a 32-core machine. That is a scheduling change and not a numerical one, and the full-precision output confirms it: every value identical to the digit.

What is calibrated, and what is predicted

This is the section that decides how much weight anything else can carry.

Calibrated. The absorber amounts are scaled by one common factor chosen to put the base state at an Earth-like surface temperature. The continuum that closes the atmospheric window is one tuned number rather than fitted MT_CKD coefficients. The cloud albedos and fraction are chosen to reproduce Earth's clear-sky and all-sky albedos. The cloud deck's height was chosen by requiring the longwave cloud effect to match observation. In the grey configurations, the CO_2 share of the absorber is tuned against a known forcing.

Predicted. Which bands exist, how opaque each is relative to the others, and how absorption is distributed within each — all of that comes from the line data. So does the *shape* of the CO_2 response: the logarithm is not imposed anywhere, and the absorber amount is exactly linear in concentration, so the curvature is produced entirely by band structure. The forcing coefficient that comes out of it is a measurement, reported as it falls.

The two dialled gases are calibrated differently, and it matters. CO_2's amount is set through the absorber scale, which is chosen against the *base state* — so its forcing coefficient falls out afterwards as a genuine prediction, 0.886× the accepted law. Methane's share is set against its *forcing* directly, so that magnitude is calibrated in and predicts nothing. What methane still predicts is its shape, and the out-of-sample agreement at 3500 ppb is the evidence that the shape is real rather than fitted.

The distinction matters most for the headline comparison. Because the CO_2 forcing is measured in W m^{-2} and compared against a law also expressed in W m^{-2}, **nothing is converted and no sensitivity is borrowed**. Turning the accepted law into a temperature would need the model's own sensitivity, which would make the reference partly a restatement of the thing it is meant to test — which is why the temperature views carry no reference curve at all.

A configuration with nothing fitted

Everything above runs on absorber amounts scaled by one common factor chosen to land the surface at an Earth-like temperature. That factor is a fit, and it absorbs every deficiency in the model at once, so none of them can be seen separately — and the surface temperature is an input rather than a result.

`EarthlikeConfiguration` removes it. Each gas gets the column density its observed abundance implies, the optical depth follows from HITRAN's own line strengths, and the surface temperature is whatever comes out. The physical magnitude was always present in the line data — `AbsorptionCoefficients` divides it away to work in shape alone — so exposing it needed one property, not new physics.

```
tau = (band-mean cross-section, cm² per molecule) × (column, per cm²)
```

285 ppm of CO_2 gives 6.12×10^{21} per cm^2, against the standard 6.1×10^{21}

	NOTHING FITTED	CALIBRATED	EARTH
Surface temperature	289.89 K	286.80 K	~288 K
Convective top	10.83 km	5.83 km	~8–11 km
CO_2 coefficient A	4.250	4.738	5.35
— against accepted	0.794×	0.886×	1.00×
Drift across the sweep	22.6 %	9.9 %	–

The unfitted configuration against the calibrated one, and against Earth

It reaches a better base state and a worse forcing. Within two kelvin of Earth's surface with no scale factor at all, and a tropopause height far closer to the real one than the calibrated configuration manages — but a CO_2 coefficient further from the accepted law, with more than double the drift.

The inventory says why. Water vapour carries **98.8 %** of the band-mean optical depth at its observed column, where the fitted recipe gave it four times CO_2 rather than eighty-five times. CO_2's band sits under far more water vapour than the fitted version, which masks its forcing and does so increasingly with concentration. The scale factor was never only setting the total — it was setting the *ratio between gases*, and that ratio is what the coefficient depends on.

Two things this configuration does not do, stated because they would otherwise be silent. The water-vapour continuum is left out rather than carried over, because the shipped continuum is a tuned number and importing it would put a fitted parameter back into a configuration whose purpose is not to have one. And the vapour is a fixed observed column rather than a Clausius–Clapeyron absorber, so there is no water vapour feedback here at all. It is a parallel configuration, not a replacement.

Results

	CLEAR SKY	67 % CLOUD
Absorber scale	17.762	5.698
Base surface temperature	286.781 K	286.793 K
Forcing at 580 ppm	3.618 W m⁻²	3.351 W m⁻²
Forcing at 1000 ppm	6.082 W m⁻²	5.852 W m⁻²
Fitted coefficient A	4.717	4.664
Against accepted 5.35	0.882×	0.872×
Drift across the sweep	10.1 %	1.2 %
Warming to 1000 ppm	+4.22 K	+3.42 K
— with vapour held fixed	+2.23 K	+2.48 K
Sensitivity	0.710 K/W m⁻²	0.586 K/W m⁻²
Convective top, 285 → 1000 ppm	5.83 → 5.83 km	5.83 → 5.83 km
Emission-temperature level	4.52 → 5.13 km	4.24 → 4.75 km

Spectral configuration, 16 bands × 16 g-points, 400 cm^{-1} wing cutoff, corrected with the published CO_2–N_2 ν_2 χ factor. A is the slope of a fit with an intercept, and drift is the spread of the per-point $F/\ln(C/C_0)$.

The vertical numbers are the mechanism made visible. Adding CO_2 lifts the height at which the column reaches the planet's emission temperature, and the surface is warmer than that level by the lapse rate times its height. That is the greenhouse argument, measured rather than asserted.

A cloud deck masks 4–7 % of the CO_2 forcing, the larger share at lower concentrations — it already blocks part of the upward flux the extra gas would have intercepted.

The clouded response is also far closer to a pure logarithm, 1.2 % drift against 10.1 %. That used to be recorded here as suggestive rather than established, because holding the base state fixed under cloud required dropping the gas loading threefold, so cloud and loading moved together. Crossing the two separates them.

	LOADING 17.762	LOADING 5.698
Clear sky	10.1 %	1.6 %
67 % cloud	10.3 %	1.2 %

Drift of the fitted coefficient: each sky at each loading

It is the loading, not the deck. Drift follows the columns and ignores the rows: the cloud's light gas load under a clear sky drifts 1.6 % with no cloud present at all, and the clear-sky load under a full deck still drifts 10.3 %. That also fits the mechanism above — less CO_2 means a less saturated band, which leaves more unsaturated wing to expand into and pushes back against χ's sub-logarithmic pull. Both low-drift cells show the coefficient rising and then falling, peaking near 500–580 ppm, which is the crossover; both high-drift cells fall throughout.

How it is checked

420 tests, and the ones worth naming are those that check the model against something outside itself.

- **Against a resolved spectrum.** The correlated-k reduction is checked against direct line-by-line integration of the same spectrum, so the quadrature is verified rather than assumed.
- **Against analytic limits.** The exponential recurrence is checked to reduce to the Koenigsberger form as $dz \to 0$; area under a broadened line is conserved; a spherical shell's emission is checked against an independent RK4 integration of the spherical two-stream equations.
- **Against observation.** The cloud radiative effect against CERES; the Bowen ratio against Earth's global mean when evaporation is enabled.
- **Convergence rather than agreement.** Grid, band count, g-points and wing cutoff are each refined and the answer's movement recorded. This is what identified the far wings as the dominant error, and what exposed an earlier configuration that had got the right coefficient by two errors cancelling.
- **Invariants that catch structural mistakes.** Forcing is zero at the reference concentration by construction, at every cloud fraction — an assertion that cost nothing and later caught a real bug, where an all-sky baseline was being differenced against a fully cloudy perturbed state.

Continuous integration builds on Windows and on Linux, the latter restoring with no package source at all, so a dependency that quietly crept in would fail the build rather than the user.

What it is not

One column, globally averaged, with no horizontal transport, no diurnal or seasonal cycle, and no dynamics. It cannot produce a circulation, a latitude gradient, or anything that depends on one.

- **The absorber amounts are illustrative.** They are scaled to reach an Earth-like surface, not taken from observed concentrations, so the CO_2 column is not Earth's. What that does to the forcing coefficient has not been isolated and is a genuine unknown inside the quoted 0.89×.

What can be measured is how saturated the band ends up. In the 634–690 cm^{-1} core the hemispheric optical depth is 2160, and *all sixteen* of its g-points are optically thick — the weakest is 89.5. Nothing in that band can respond to more CO_2; the whole forcing comes from the flanking bands and the far wings, which is why the wing treatment dominated everything else.

Whether that is *more* saturated than the real atmosphere is not established here. It would need a reference line-by-line calculation the model does not do, and the derivation's amount units are arbitrary, so the scale factor cannot answer it either.

- **Line strengths are not temperature-scaled.** Half-widths are, through HITRAN's n_{air}, but intensities are used at their 296 K values because scaling them needs total internal partition sums. Line mixing, Voigt profiles, self broadening and pressure shift are all absent.
- **The instantaneous forcing is not the adjusted forcing.** The accepted 5.35 $\ln(C/C_0)$ is a stratosphere-adjusted value at the tropopause; this is an instantaneous one at the top of the atmosphere. They are different quantities, and the comparison should be read with that in mind.
- **One grey cloud deck** at one height stands in for everything from fog to cirrus, and is prescribed rather than predicted — so it cannot act as a feedback, which is the single largest uncertainty in real sensitivity estimates.
- **The shortwave is a deposition profile, not a transfer calculation.** There is no scattering anywhere in the model; cloud and surface reflection are albedos.
- **No feedbacks except water vapour.** No ice–albedo, and no lapse-rate feedback beyond what the fixed critical lapse rate already imposes.

What it is for is showing that the qualitative behaviour of a radiative–convective atmosphere — a convecting troposphere under a radiative stratosphere, a greenhouse effect that scales with the height of the emission level, and a logarithmic response to CO_2 — follows from a small number of physical statements applied carefully, and can be traced from HITRAN line data to a surface temperature without any step being taken on trust.

18,451 lines of C# in 69 files — 6,190 of physics in `ClimateColumn.Core`, 39 % of it comment, and 8,900 lines of tests. 420 tests pass. Longwave opacity is derived from HITRAN line data, which is not redistributed with the source.

`github.com/MarkAndersUK/ClimateColumn`